

The Coverage Illusion: From Pre-retrieval Routing Failure to Post-retrieval Cascades in a Production RAG System

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Abstract

In modern RAG pipelines, query augmentation methods such as HyDE and query expansion are applied to every query, resulting in substantial LLM inference costs and increased end-to-end latency. The empirical justification for this overhead in real production traffic remains largely unexplored. We present a case study of the Danish National Encyclopedia, evaluating five retrieval workflows over 20,000 query–workflow pairs from production traffic and synthetic conditions. In this system, synthetic queries suggest that LLM augmentation is needed for over 90% of queries to achieve high retrieval coverage. However, under our production deferral policy, only 27.8% of real user queries need LLM augmentation. We call this gap the *Coverage Illusion* and attribute it to a structural mismatch between synthetic and real query distributions. Pre-retrieval routing cannot resolve this gap, as the need for LLM augmentation is only revealed after searching the index, a result confirmed by our evaluation of four machine learning paradigms. The coverage gap, undetectable from the query alone, motivates a post-retrieval cascade that runs workflows in cheapest-first order and escalates to LLM augmentation only when a step returns no documents. Operating entirely without training overhead or secondary serving infrastructure, the cascade improves quality by +0.140 Composite Overall points over Always-HyDE, reduces latency by 31.8%, and serves 72.2% of real user queries without LLM augmentation.

1 Introduction

Modern Retrieval-Augmented Generation (RAG) systems increasingly use Large Language Models (LLMs) not just for generation but within the retrieval process itself. Techniques such as HyDE (Gao et al., 2022) and query expansion (Jagerman et al., 2023; Ma et al., 2023) rewrite or enrich queries before the index is searched, often yielding substantial gains in retrieval coverage but at a high

cost in compute and latency. Despite this overhead, both are applied uniformly to every query in production systems (Shen et al., 2024b; Jeong et al., 2024), with no per-query assessment of whether augmentation is necessary.

We study the Danish National Encyclopedia, a production RAG system serving real users across a corpus of ~240,000 curated articles. Two features of this deployment make the question of uniform LLM augmentation consequential. The system defers rather than speculates when retrieval returns no relevant documents, so retrieval *coverage* is the dominant quality factor. Furthermore, real users submit short keyword phrases, quite unlike the verbose conversational queries that dominate English RAG benchmarks (Fan et al., 2024). Both features motivate the question of whether LLM augmentation can be applied selectively rather than uniformly. A natural solution is pre-retrieval query routing, in which a lightweight classifier assigns each query to the most likely successful workflow (Ong et al., 2025; Zhang et al., 2026). However, we demonstrate that pre-retrieval routing fails because the need for LLM augmentation depends on the retrieval index rather than the query itself. Our analysis yields three main findings.

- 1. The Coverage Illusion.** Our synthetic queries indicate LLM augmentation is necessary for over 90% of queries. On real user queries from the same deployment, only 27.8% require LLM augmentation under our deferral policy, as retrieval-only Hybrid already returns sources for the remaining 72.2%.
- 2. Pre-retrieval Routing Fails.** Across four Machine Learning (ML) paradigms (rule-based heuristics, classification, neural fine-tuning, and regression), every approach fails on real user queries. In an entity-heavy reference corpus, whether a query needs augmentation is a property of the index, not of the query text.

3. **Post-retrieval Cascade.** Since the need for augmentation is revealed only after retrieval runs, the natural response is a cascade in which retrieval proceeds in cheapest-first order and escalates only when a step returns no documents. The post-retrieval cascade improves quality by +0.140 Composite Overall points over the best static baseline, reduces latency by 31.8%, and invokes LLM augmentation for only 27.8% of queries.

2 Related Work

Synthetic vs. Real Query Evaluation. Synthetic queries have become the standard benchmarking tool for RAG systems (Santillán Cooper et al., 2025; Jadon et al., 2025; Barbiero et al., 2025; Teixeira de Lima et al., 2025). This benchmarking practice introduces a fundamental structural mismatch. Synthetic queries tend to be contextually verbose and grammatically complete, whereas real user interactions are characterized by brevity and extreme vocabulary sparsity (van Elburg et al., 2025; Rahmani et al., 2025; Kim and Baek, 2025). How far this gap skews system-level conclusions, particularly about when retrieval augmentation is necessary, remains unanswered (Thakur et al., 2021).

Production RAG Systems and Case Studies. Despite increasing research on RAG evaluation and adaptive routing (Sharma, 2025; Teixeira de Lima et al., 2025), empirical analyses grounded in production settings remain relatively rare. Most studies rely on benchmark datasets that may not reflect real-world query distributions (Simon et al., 2024; Romero Lauro et al., 2026). Our work addresses this gap by analyzing a production system in which query distributions, retrieval characteristics, and quality incentives are drawn from live traffic, enabling us to measure how far synthetic evaluation diverges from real deployment behavior.

Query Routing and Adaptive RAG. Adaptive RAG systems are designed to steer each query toward the most appropriate retrieval path. Frameworks like LangChain (LangChain, 2023) and LlamaIndex (LlamaIndex, 2023) use hand-crafted heuristics; Gorilla (Patil et al., 2024) and RouteLLM (Ong et al., 2024) learn routing policies from data; Self-RAG (Asai et al., 2023) decides dynamically whether to retrieve at all. All these systems commit to a routing decision *before* retrieval, based solely on the input query (Varangot-Reille

et al., 2025). To our knowledge, no prior work has questioned whether the query alone provides sufficient signal for this, nor explored routing conditioned on what retrieval returns (Sharma, 2025).

Advanced Retrieval Strategies for RAG. HyDE (Gao et al., 2022) and LLM-based query expansion (Han et al., 2020; Gao et al., 2021; Ma et al., 2023; Jagerman et al., 2023) improve retrieval for underspecified queries by rewriting or enriching them. Both methods add substantial inference overhead (Lewis et al., 2020; Shen et al., 2024a), yet most prior work applies LLM augmentation uniformly to all queries, treating it as universally beneficial (Chen et al., 2023).

3 Experimental Setup

Our testbed is the Danish National Encyclopedia (lex.dk), a corpus of approximately 240,000 professionally curated articles, semantically chunked, embedded with multilingual-e5-large (Wang et al., 2024), and indexed in a vector database. Three properties suit this deployment for the study. It carries real production traffic, its entity-heavy content produces clear retrieval outcomes, and its stable curation ensures performance differences are attributable to the retrieval workflow rather than corpus noise. Five retrieval workflows operate on this index under four query conditions of 1,000 queries each; each pair is scored by an automated LLM judge, yielding 20,000 evaluated pairs (Figure 4, Appendix). From these, we construct an oracle dataset that identifies the best-performing workflow for each query, which serves as the benchmark for pre-retrieval routing and the cascade.

3.1 Query Conditions

We evaluated four query conditions, each with 1,000 queries. One condition draws from production logs (Real User), while the remaining three are synthetic variants spanning a range of formulation styles, from raw LLM output to natural conversational language (Synth-Polluted, Synth-Keywords, and Synth-Conversational). Full details of each condition, including sources and query statistics, are provided in Appendix B (Tables 6–7).

Real User queries come from production search logs and are the primary reference condition for all deployment-relevant conclusions. They have a median length of 5.5 words, typically consist of a bare topic fragment or keyword phrase, as in *Menneskets anatomi* ('Human anatomy'). Users name a topic

rather than describe it, a pattern that maps naturally to BM25 (Robertson and Zaragoza, 2009) lexical matching but gives dense retrieval little semantic signal to work with. Their brevity and lexical sparsity make them the most challenging condition for dense retrieval and the only representative of live production traffic. A further 5,000 queries drawn from the same production logs are held out from all routing experiments and used exclusively for the generalization tests in pre-retrieval routing.

Synth-Polluted queries are the raw output of GPT-5.2 (OpenAI), generated from article content using a structured prompt. The outputs are instruction-heavy and chatty, as in *Svar ja eller nej og begrund kort: adskiller impressionismen sig fundamentalt fra tidligere europæiske malestile?* ('Answer yes/no and briefly justify: does impressionism differ fundamentally from earlier European painting styles?'). The remaining two synthetic conditions rewrite these queries (see Appendix A).

Synth-Keywords queries are created by rewriting each Synth-Polluted query into a short keyword phrase using GPT-3.5 Turbo (Developers), as in *impressionisme fransk maleri 1800-tal stil* ('impressionism French painting 19th-century style'). They are shorter than Synth-Polluted queries, but more vocabulary-rich than real user queries.

Synth-Conversational queries are created by rewriting each Synth-Polluted query into a natural-language question using GPT-3.5 Turbo (Developers), as in *Hvad adskiller impressionismen fra tidligere europæiske maleritraditioner?* ('What distinguishes impressionism from earlier European painting traditions?'). They closely resemble the query style used in most RAG benchmarks and represent an optimistic setting for LLM-augmented retrieval, as they are rich, contextually complete, and well aligned with the corpus semantics.

3.2 Retrieval Workflows

The five workflows span the full cost spectrum and are organized into two tiers (illustrated in Figure 5, Appendix). Tier 1 retrieves directly from the vector index; Tier 2 first transforms the query by invoking an LLM, and then retrieves.

Tier 1 workflows retrieve directly from the vector index. **Semantic** embeds the raw query and retrieves the top- k ($k = 10$) nearest neighbors from the vector index. **Semantic-CE** does the same but adds a Danish cross-encoder reranker (MiniLM-L6-danish-reranker-v2 (Martinsen)) to rescore the top- k candidates. **Hybrid** combines BM25 full-

text search with dense vector retrieval, merging the top- k ranked lists from both sources via Reciprocal Rank Fusion (RRF) (Cormack et al., 2009).

Tier 2 workflows transform the query with an LLM before retrieval. **QE-CE** prompts an LLM to expand the query with related terms, then retrieves with a hybrid search and reranks with the cross-encoder reranker (Martinsen). **HyDE** prompts an LLM to generate a hypothetical answer document, which is then used for retrieval via its embedding.

3.3 Evaluation Protocol

Scale and Parameters. The full evaluation covers 20,000 query-workflow pairs, comprising five retrieval workflows evaluated across four query conditions of 1,000 queries each. Responses are generated by gemma-4-26B-A4B-it (Google) with default parameters; quality scores come from Qwen3-32B (Yang et al., 2025) acting as an automated LLM judge, running greedy decoding (temperature=0.0, max_tokens=4096) for deterministic, reproducible evaluations. All reported latency figures are end-to-end wall time, including LLM generation. To check that our findings do not depend on the specific judge, we replicated the evaluation with Qwen3.6-27B (Yang et al., 2025) (comparison of both given in Appendix F).

Metrics. Each response receives three scores from the automated judge on a 1–5 scale, covering faithfulness (claims grounded in retrieved sources), answer relevance (response addresses the question), and retrieval quality (documents are on-topic). Our primary metric, `composite_overall` (**CO**), averages faithfulness and answer relevance across all queries; retrieval quality is reported separately as a process diagnostic. Queries returning no documents are assigned `CO = 1.0`, the minimum, since the system must defer rather than generate an ungrounded response. We also report `composite_when_answered` (**CWA**), computed only over answered queries, which isolates per-answer quality from the coverage effect. The binary `has_sources` flag (1 for answered queries and 0 for deferred ones) partitions these two groups and serves as the escalation trigger in the cascade.

4 The Coverage Illusion

We establish per-workflow baselines on all 1,000 real user queries. Comparison with synthetic conditions then reveals how far standard benchmarks overstate the need for LLM augmentation.

4.1 Static Baselines

Table 1 reports results when each workflow is applied uniformly to all 1,000 real user queries. CO follows coverage closely across all five systems, from Semantic (62.6%, CO=3.058) up to HyDE (86.4%, CO=3.944), confirming that coverage is the primary driver of composite quality. The narrow CWA spread (4.287–4.407, a range of just 0.12 points) shows that once retrieval succeeds, all five workflows produce comparably good answers. Always-HyDE incurs 96.2s of end-to-end latency, 2.6× more than the Hybrid baseline (37s), without a commensurate gain in answer quality. The oracle ceiling (CO=4.404) marks the best achievable quality under perfect per-query routing.

Workflow	Method	CO	CWA	Cov.	Lat.
<i>Tier 1: Retrieval-Only</i>					
Semantic	Dense vector search	3.058	4.287	62.6%	33s
Semantic-CE	Dense + CE rerank	3.086	4.321	62.8%	33s
Hybrid	BM25 + Dense + RRF	3.408	4.335	72.2%	37s
<i>Tier 2: LLM-Augmented</i>					
QE-CE	LLM expansion + CE rerank	3.601	4.338	77.9%	60s
HyDE	Hypothetical doc. embedding	3.944	4.407	86.4%	96s
Oracle	—	4.404	4.940	—	—

Table 1: Retrieval workflows and static performance; latency is end-to-end wall time, including generation.

4.2 Coverage Drives Quality

LLM-augmented workflows score higher because they retrieve sources more frequently, not because they generate better answers. The binary has_sources flag explains 69–83% of overall quality variance (mean $R^2=0.77$; Appendix G). To quantify this, we decompose the difference in composite overall score (ΔCO) between HyDE and the Semantic baseline using a first-order approximation. The decomposition is defined as:

$$\Delta\text{CO} \approx \Delta\text{cov} \times \bar{Q}_{\text{baseline}} + \text{cov}_{\text{baseline}} \times \Delta\bar{Q}_{\text{ans}} + \epsilon$$

where Δcov is the coverage gain, $\bar{Q}_{\text{baseline}}$ is the baseline’s mean per-answer quality, $\text{cov}_{\text{baseline}}$ is the baseline’s coverage rate, $\Delta\bar{Q}_{\text{ans}}$ is the improvement in per-answer quality, and ϵ is a residual term that captures the marginal performance of queries that are resolved only by the augmented workflow. Under this formalization, coverage expansion contributes +1.020 CO points and improved synthesis contributes +0.076 CO points. The sum of these two components exceeds the observed total because the residual term $\epsilon = -0.209$ is negative. This occurs because the queries HyDE reaches

that Semantic does not tend to be harder, and their actual per-answer quality falls below Semantic’s average, causing the approximation to overshoot. HyDE’s advantage is almost entirely due to retrieving sources that Semantic misses, with minimal impact from synthesis quality (Appendix H).

4.3 The Gap Between Synthetic and Real

On synthetic queries, LLM-augmented workflows achieve 93–95% coverage, far ahead of retrieval-only alternatives; on real user queries, this advantage collapses. Retrieval-only Hybrid already attains 72.2% coverage, and HyDE raises it only to 86.4% (a 14.2 point gain), versus a 32.9 point HyDE–Hybrid gap on Synth-Keywords. This disparity is the Coverage Illusion in practice (Figure 1). In our evaluation, the synthetic query conditions imply augmentation is necessary for over 90% of queries, yet on real user queries from the same deployment, only 27.8% require escalation under our deferral policy. The underlying cause is a structural difference between query types. Real user queries have a mean length of 7 words, typically a keyword phrase that BM25 maps directly to the relevant article. Synthetic queries average 10–20 words and use broader, rich vocabulary, which reduces lexical overlap with the corpus and makes retrieval-only approaches appear less effective than they are on real traffic.

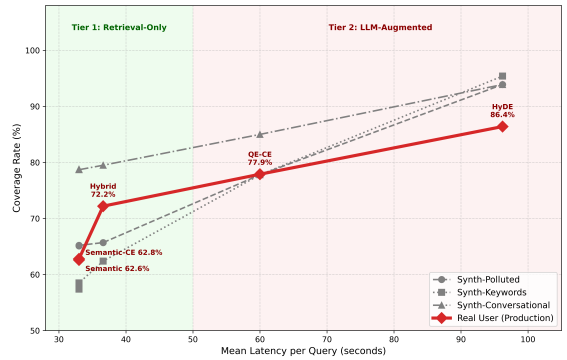


Figure 1: The Coverage Illusion, showing retrieval coverage by workflow and query condition.

5 The Limits of Pre-retrieval Routing

If 72.2% of real user queries do not require LLM augmentation, a lightweight pre-retrieval classifier could, in principle, identify those queries in advance and skip the LLM augmentation step entirely. We tested this across four machine learning paradigms of increasing complexity. To fo-

cus on queries where the workflow choice actually matters, we evaluated on a subset of 2,206 oracle queries where the best and worst workflow diverge by at least 1.5 CO points; among these, 66.4% have an LLM-augmented oracle label and 33.6% a retrieval-only label (Table 8). For real user queries specifically, 609 of 1,000 meet this threshold, split nearly evenly at 52.4% LLM-augmented and 47.6% retrieval-only. We held out 20% of the dataset as a stratified test set for all ML experiments, giving $n = 442$ for classification and regression models; the rule-based heuristics use a slightly larger split ($n = 494$) because oracle labels are recomputed with a looser tie-breaking threshold.

Since the high-contrast subset filters out queries where workflows tie, Always-HyDE on this subset scores CO=4.258, above the full-set figure of 3.944 in Table 1; queries where HyDE offers no advantage over cheaper workflows are excluded. Against this elevated baseline, every routing model fell well short of the oracle ceiling, and all models degraded further when evaluated on real user queries.

5.1 Rule-Based Heuristic Routing

Handcrafted string rules (query length, question marks, Danish question words) reach **38.4% accuracy**, just above the uniform-random baseline of 20% for five classes. The heuristic routes queries to three of the five workflow classes; QE-CE and Semantic-CE receive zero correct predictions. Per-class recall is poor across the covered workflows (Semantic: 63.2%, HyDE: 37.3%, Hybrid: 25.8%). On the stratified test split ($n = 494$), the router achieves CO=4.202, **below** Always-HyDE (4.258).

5.2 Classification Routing Models

Table 2 shows that only the simplest feature configurations outperform Always-HyDE (CO=4.258). Logistic Regression (LR) with TF-IDF character n-grams achieves the best result (CO=4.269, $\Delta = +0.011$), while LR with surface features (character counts, question marks, stop-word ratios) follows closely (CO=4.260, $\Delta = +0.002$). LR with word n-grams falls below the baseline (CO=4.240, $\Delta = -0.018$). Even the best model recovers just 2.4% of the 0.475-point gap between Always-HyDE and the oracle ceiling.

Richer features and greater model capacity consistently worsen routing quality. A Gradient Boosting Machine (GBM) with TF-IDF and surface features scores CO=4.177 ($\Delta = -0.081$), below every LR variant. A GBM over mul-

Model	CO	Δ vs HyDE
LR / TF-IDF char (3–5)grams	4.269	+0.011
LR / surface features	4.260	+0.002
LR / TF-IDF word (1–2)grams	4.240	−0.018
GBM / TF-IDF + surface	4.177	−0.081
GBM / multilingual embed.	4.184	−0.074
LR / multilingual embed.	3.850	−0.408
Always-HyDE (baseline)	4.258	—
Oracle (ceiling)	4.733	+0.475

Table 2: Classification model router performance on oracle test split ($n = 442$, high-contrast queries only).

tilingual embeddings reaches only CO=4.184 ($\Delta = -0.074$). LR over multilingual embeddings (paraphrase-multilingual-MiniLM-L12-v2) (Reimers and Gurevych, 2019) is the worst configuration (CO=3.850, $\Delta = -0.408$); minority classes such as QE-CE (21% recall) and Semantic-CE (29% recall) are predicted nearly at random. Across all six models, richer representations yield less routing signal than character n-grams because multilingual embeddings are designed to suppress surface differences in favor of semantic similarity, removing the surface variation that weakly tracks whether a query will match the index by keyword.

On 5,000 unseen real user queries, the best LR model (TF-IDF character n-grams) routes over 90% of queries to HyDE, capturing under 5% of the augmentation-free opportunity versus the 33.6% oracle ceiling (Table 8). Even for one-to-three-word queries, where BM25 coverage is highest, the most are still routed to HyDE, confirming that query length is not a reliable routing signal. Retraining on the 609 real user queries reduces domain shift, but augmentation-free routing reaches only **13.4%** (of the oracle ceiling), limited to the two cheapest workflows. The cascade (Section 6) achieves 72.2% augmentation-free routing with no training, $5.4\times$ rate of the best ML router.

5.3 Neural Encoder Fine-Tuning

We fine-tuned all-MiniLM-L6-v2 (Reimers and Gurevych, 2019) for 5 epochs on the oracle labels. Evaluated on 5,000 blind real user queries, the model routed 73.8% of queries to HyDE and 26.2% to Hybrid, assigning none to the other workflows. Training loss fell by 16% (1.224→1.025) over 5 epochs, pointing to signal failure rather than insufficient training. Weighting CO by routing share yields an estimated CO of **3.804**, below Always-HyDE. The model separates broad augmentation tiers but cannot discriminate within a tier.

5.4 Regression-Based Routing Models

We trained a regression model to predict the quality score for each workflow based on the query text. The system defaults to HyDE; a cheaper alternative replaces it only when HyDE’s predicted score exceeds the Alternative’s by at least δ points, a threshold that guards against low-confidence substitutions. Table 17 (Appendix J) reports the Pareto frontier for Ridge regression under 5-fold cross-validation.

The best operating point ($\delta = 0.75$) captures only **3.7%** of the oracle gap, redirecting just 4.1% of queries away from HyDE, well short of the 33.6% retrieval-only ceiling (Table 8; Appendix C). GBM regression with far greater capacity performs worse (CO=4.222, $\delta = 2.00$, capturing 1.4% of the oracle gap), confirming that the bottleneck is the missing signal, not model complexity. Above $\delta = 0.75$, the router collapses to Always-HyDE. Below $\delta = 0.50$, quality falls below the HyDE baseline, as the low threshold causes the model to route too many queries away from HyDE, including queries that genuinely needed it.

5.5 Root Cause: Query Content Limitations

All four paradigms fail for the same reason. The need for LLM augmentation is determined by what the retrieval index contains, not by what the query says. Whether a given query needs augmentation depends on whether the index holds directly relevant content for it, which is unknowable before retrieval runs. No surface feature, embedding, or learned classifier can substitute for this post-retrieval signal. We hypothesize that this limitation generalizes to any entity-heavy reference corpus where real users submit direct factual lookups.

6 Post-Retrieval Cascade Architecture

6.1 Design

The failure of pre-retrieval routing demonstrates that query-side features cannot anticipate the contents of the retrieval index, making pre-retrieval prediction inherently limited. The cascade instead responds to the actual retrieval outcome, running workflows in order of increasing cost and escalating only when a step returns no documents. Its value is empirical rather than architectural, quantifying the margin by which a reactive policy outperforms any trained routing alternative. As shown in Figure 2, if retrieval succeeds the LLM generates an answer

and the process stops; otherwise, the cascade advances to more expensive workflows, ultimately deferring if no evidence is found.

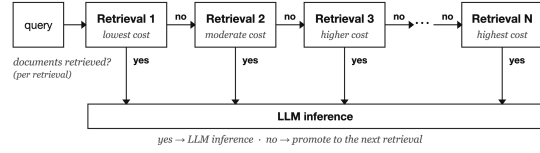


Figure 2: Post-retrieval cascade execution flow.

Formally, let q be the incoming query and $\mathcal{W} = (w_1, w_2, \dots, w_n)$ the ordered sequence of retrieval workflows, sorted by ascending cost. For each $w_i \in \mathcal{W}$, let $R(q, w_i)$ denote the retrieved document set.

The cascade evaluates a binary success indicator $I(q, w_i)$:

$$I(q, w_i) = \begin{cases} 1 & \text{if } |R(q, w_i)| > 0 \\ 0 & \text{otherwise} \end{cases}$$

If $I(q, w_i) = 1$, the cascade terminates, and the retrieved document set $R(q, w_i)$ is passed to the generation model. If $I(q, w_i) = 0$, the system escalates, initiating retrieval for the next workflow w_{i+1} in the sequence. If all workflows return empty results, the system applies the deferral policy.

6.2 Efficiency of the Stopping Condition

The `has_sources` check is an $O(1)$ array operation on the retriever’s return value, as no LLM is invoked at this stage. The binary threshold is also empirically optimal. To confirm this, we ran offline simulations with relaxed escalation thresholds. Escalating when fewer than two or three documents were retrieved, rather than when none were found, reduced composite quality from 4.03 to 3.98. When at least one well-matched document is already available, a hypothetical document adds noise rather than coverage.

7 Results

7.1 Cascade Performance: Real User Queries

Across all 1,000 real user queries, the cascade improves quality, cuts latency, and limits LLM augmentation to 27.8% of queries, compared to 100% under Always-HyDE (Table 3). Composite score rises from CO=3.944 to CO=4.084 ($\Delta = +0.140$), end-to-end latency falls 31.8% from 96.2s to 65.6s, and the language model is invoked only for queries where Hybrid retrieval fails. These gains do not

trade quality for speed. 72.2% of real user queries stop at the first cascade step and never incur the LLM transformation that Always-HyDE applies to all queries. The lower Always-HyDE baseline of CO=3.944 here, compared with CO=4.258 in Section 5, reflects the inclusion of the full query set, which restores tied queries for which HyDE offers no advantage over cheaper workflows.

Strategy	CO	Δ CO	Latency	Saved	Aug.%
Always-HyDE	3.944	—	96.2s	—	100%
Cascade	4.084	+0.140	65.6s	31.8%	27.8%
Oracle ceiling	4.404	+0.460	—	—	—

Table 3: Best cascade vs. Always-HyDE on all 1,000 real user queries.

27.8% of all queries for which Hybrid returns no documents escalate to QE-CE or HyDE. The quality gain over Always-HyDE has two sources. First, the cascade avoids the quality degradation that HyDE causes on queries that keyword matching resolves cleanly. When the relevant article is a direct lexical match, HyDE’s hypothetical document redirects retrieval toward adjacent but less relevant sources (the per-query breakdown is in Section 8). Second, the cascade reserves HyDE’s coverage for queries that genuinely need it.

Table 4 compares all tested cascade orderings. The three-step configuration (Hybrid $\rightarrow$ QE-CE $\rightarrow$ HyDE) achieves the highest quality (CO=4.084). The two-step variant (Hybrid $\rightarrow$ HyDE) saves an additional 2.8 seconds at a cost of only 0.016 CO points; inserting Semantic-CE as an intermediate step incurs additional latency without improving quality. The cascade outperforms Always-HyDE in every ordering tested, confirming the gains are robust to the specific steps chosen. The oracle ceiling (CO=4.404) marks the best achievable quality under perfect per-query routing; every pre-retrieval method we tested covered at most 2.2% of this gap (Table 5), whereas the cascade, without training or additional infrastructure, covers 30%.

Cascade	CO	Δ CO	Lat.	Saved
Hyb$\rightarrow$QE-CE$\rightarrow$HyDE	4.084	+0.140	65.6s	31.8%
Hyb $\rightarrow$ Sem-CE $\rightarrow$ QE-CE $\rightarrow$ HyDE	4.078	+0.134	68.6s	28.7%
Hyb $\rightarrow$ HyDE	4.068	+0.124	62.8s	34.7%
Sem $\rightarrow$ Hyb $\rightarrow$ QE-CE $\rightarrow$ HyDE	4.050	+0.106	68.9s	28.4%
Always-HyDE	3.944	—	96.2s	—

Table 4: Cascade variants on real user queries.

The +0.140 advantage is observed across all 1,000 real user queries without filtering. Restrict-

ing to the 609 real user queries that form the high-contrast routing subset (where workflows diverge by at least 1.5 CO points), the advantage grows to $\Delta = +0.241$ (CO=3.936 vs. HyDE CO=3.695; Table 19). The routing subset retains only queries where the workflow choice affects quality.

7.2 Pareto Dominance

Figure 3 shows that the cascade is the only point combining strictly higher quality with strictly lower cost. The 30.6-second saving is real computation avoided, as 72.2% of queries stop at Hybrid and never invoke LLM augmentation. Among the static baselines, Hybrid offers comparable speed but fails to retrieve sources for 27.8% of queries; the cascade gives Hybrid’s speed to the majority while reserving HyDE for the minority that needs it.

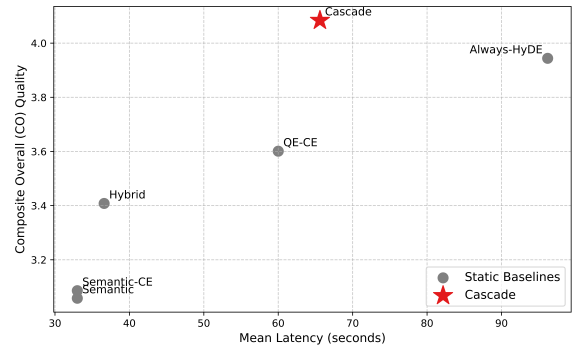


Figure 3: Cost vs. Quality on real user queries.

7.3 Cross-Condition Analysis

Table 18 (Appendix K) evaluates the cascade across all four query conditions. On real user queries, the cascade wins by +0.140 CO with 31.8% latency savings. Even on Synth-Conversational queries, the condition most favorable to LLM augmentation, the cascade matches HyDE (+0.010 CO) while saving 41.9% in latency. On Synth-Keywords and Synth-Polluted queries, Always-HyDE outscores the cascade by 0.139 and 0.048 CO points respectively, though the cascade still saves roughly 29–30% in latency in both cases.

This reversal is the Coverage Illusion in action. The condition under which HyDE wins, rich and verbose queries, is precisely what synthetic benchmarks over-represent. Even Synth-Conversational, the most linguistically complete condition, yields only a +0.010 CO cascade advantage, confirming that vocabulary-index alignment, rather than query length or grammatical form, determines whether augmentation is needed.

7.4 Human Evaluation

Three independent annotators scored 100 stratified query–response pairs on faithfulness, answer relevance, and retrieval quality using the same 1–5 scale as the automated judge. The automated judge slightly overestimates human scores across all three metrics ($\Delta = +0.04$ to $+0.14$ points), with Spearman correlations ranging from $\rho=0.59$ to $\rho=0.78$ (Table 9, Appendix D). Workflow rankings are consistent between human and automated evaluation, confirming that the cascade’s advantage over Always-HyDE is not an artifact of the automated judge. This is corroborated by a second LLM judge, Qwen3.6–27B, which replicates the workflow ranking and coverage-dominance pattern of the primary Qwen3–32B judge (Table 11, Appendix F).

8 Discussion

The cascade’s advantage over Always-HyDE is not merely a matter of efficiency. For a significant proportion of queries, the uniform application of HyDE actively degrades response quality. Among the 722 real user queries where Hybrid retrieved sources, the per-query comparison reveals three distinct outcomes. In 59.6% of cases, the two workflows scored the same, indicating that LLM augmentation added nothing. In 21.2% of cases, Hybrid outperformed HyDE by a mean of 2.010 CO points. HyDE prevailed in only 19.3% of cases, with an average margin of 1.324 CO points.

When a direct lexical match successfully locates the target article, the language model’s hypothetical document introduces semantic drift, steering retrieval toward adjacent, less relevant passages rather than the precise target source. The result is a quality penalty. HyDE’s per-query losses in this regime (mean 2.010 CO points) systematically exceed its gains (mean 1.324 CO points), even though this asymmetry is invisible in aggregate metrics because it affects only the minority of queries where Hybrid already succeeds (Appendix L).

These findings challenge the assumption that more inference always improves retrieval. In our encyclopedia deployment, we believe that, as in other entity-heavy reference systems, most queries are simple lookups that keyword matching can handle. Routing every query through LLM augmentation is inefficient, since Hybrid alone retrieves sources for 72.2% of queries, and in those, HyDE provides no improvement in 59.6% of cases and actually reduces quality in 21.2% (Appendix L).

Strategy	Aug-free%	CO	Training
Always-HyDE (baseline)	0.0%	3.944	None
ML Router (all conditions)	<5.0%	3.954	1,764 labels
ML Router (real user)	13.4%	3.920	609 labels
Neural Router (fine-tuned)	26.2%	3.804	1,764 labels
Cascade	72.2%	4.084	None
Oracle (ceiling)	33.6%	4.404	Perfect labels

Table 5: Pre-retrieval routing vs. cascade

Table 5 makes the contrast concrete. An ML router requires oracle labels, model training, and serving infrastructure, and reaches at most 13.4% augmentation-free routing. The cascade requires only a Boolean check on the retriever’s output, and reaches 72.2% augmentation-free routing at a higher quality score than any ML alternative. Its augmentation-free rate (72.2%) exceeds the 33.6% retrieval-only oracle ceiling because that ceiling is computed on high-contrast queries; beyond that subset, Hybrid still finds relevant sources for many queries where no workflow has a strong advantage.

9 Conclusion

We document the *Coverage Illusion* in a production RAG system. Synthetic query conditions suggest that LLM augmentation is necessary for the majority of queries; on real user queries from the same deployment, retrieval-only Hybrid already returns sources for 72.2%, so only 27.8% require escalation under our deferral policy.

No pre-retrieval routing method bridges this gap. Across all four ML paradigms tested, the failure is consistent. Whether augmentation is needed for a given query depends on the retrieval index, which becomes apparent once retrieval has run.

The cascade reactively routes queries with an $O(1)$ check on the retriever’s output, requiring no training or serving infrastructure. For real user queries, it improves quality by +0.140 CO points and reduces latency by 31.8% compared to Always-HyDE, using LLM augmentation for just 27.8% of queries. The design is workflow-agnostic and adaptable to any ordered set with a binary escalation condition. RAG evaluation benchmarks dominated by synthetic queries risk overstating how often LLM augmentation is necessary, particularly for deployments where real users submit short keyword lookups against an entity-rich index. Practitioners should validate the augmentation need on real user traffic before assuming benchmark results carry over to their production setting.

10 Limitations

Single Corpus and Deployment Setting. All experiments were conducted on a single encyclopedia. The most consequential constraint on generalization is the deferral policy, under which failed retrievals receive CO=1.0 (Section 3.3), making coverage the dominant quality factor by construction. Permissive systems that fall back on parametric generation face a smaller coverage penalty, so augmentation would appear more valuable there.

Three further constraints are deployment-specific. Real encyclopedia users submit short keyword phrases at rates that deviate from standard benchmarks (40.4% keyword-style, 21.0% well-formed), making hybrid retrieval inherently competitive; conversational or long-form settings may present a different picture. Danish is also morphologically rich, and BM25 lexical matching may differ from English in ways that affect Tier 1 vs. Tier 2 performance; we make no claims about other languages.

Cascade Worst-Case Overhead. Of the 27.8% of queries that Hybrid does not resolve, approximately 12.3% stop at QE-CE, while the remaining 15.5% escalate through all cascade steps to HyDE. That 15.5% face a mean cumulative latency of ~ 161 s, because each intermediate step contributes wall time even when it returns no documents. In production, where failed steps skip generation entirely (incurring only retrieval time for Hybrid, and LLM expansion plus retrieval for QE-CE), this worst case drops to approximately 125s ($\sim 1.3\times$ the 96.2s HyDE baseline).

LLM-as-Judge Bias. All primary quality scores were produced using an LLM acting as a judge. While automated judges are widely adopted to improve scalability, they can exhibit systematic preferences for longer outputs and for stylistic alignment. We address this by conducting a human validation study with three independent annotators scoring 100 stratified query–response pairs. Spearman correlations between human and automated scores range from $\rho=0.59$ (faithfulness) to $\rho=0.78$ (answer relevance), with a mean absolute error of 0.17–0.21 points on the 1–5 scale. The LLM judge overestimates by at most 0.14 points. Crucially, workflow rankings are consistent across both evaluation paradigms, confirming that the dominance of the binary coverage signal is an intrinsic system property rather than an evaluation artifact.

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A Query Generation Prompts

The following prompts produced the three synthetic query conditions; all derive from the same 1,000 article-grounded query–answer pairs and differ only in how the query was reformulated.

A.1 Synth-Polluted - Generation Prompt for GPT-5.2

Synth-Polluted queries are the *raw output* of GPT-5.2. Each call received a full article, a randomly sampled query style from a fixed taxonomy, and a set of real production queries as stylistic anchors. The prompt was built around the following five components.

Role & context. Senior RAG evaluation data scientist at a national encyclopedia; the objective is to build an evaluation dataset that tests retrieval accuracy, multi-section synthesis, and grounded generation.

Query constraints. Each query must be (i) fully self-contained and interpretable in isolation, (ii) written in Danish, (iii) grounded strictly in the provided article, (iv) free of answer hints and article self-references, and (v) independent of all other queries in the batch.

Complexity annotation. Each query was labeled along three dimensions (*information_composition*, *article_coverage*, and *reasoning_load*).

Structure & information types. Two template variables, *structure_instructions* and *info_type_instructions*, injected a fixed taxonomy of query structures (e.g., yes/no, list, comparison) and information types (e.g., causal, temporal). The model was instructed *not* to force a type the article did not support.

Output format. A JSON list of objects, each containing query, answer, *sections_for_answer*, and a tags block holding *info_types*, styles, structures, and the complexity annotation.

A.2 Synth-Keywords - Keyword Abstraction Prompt for GPT-3.5 Turbo

Each Synth-Polluted query was rewritten into a keyword query of 5-10 words using the following prompt.

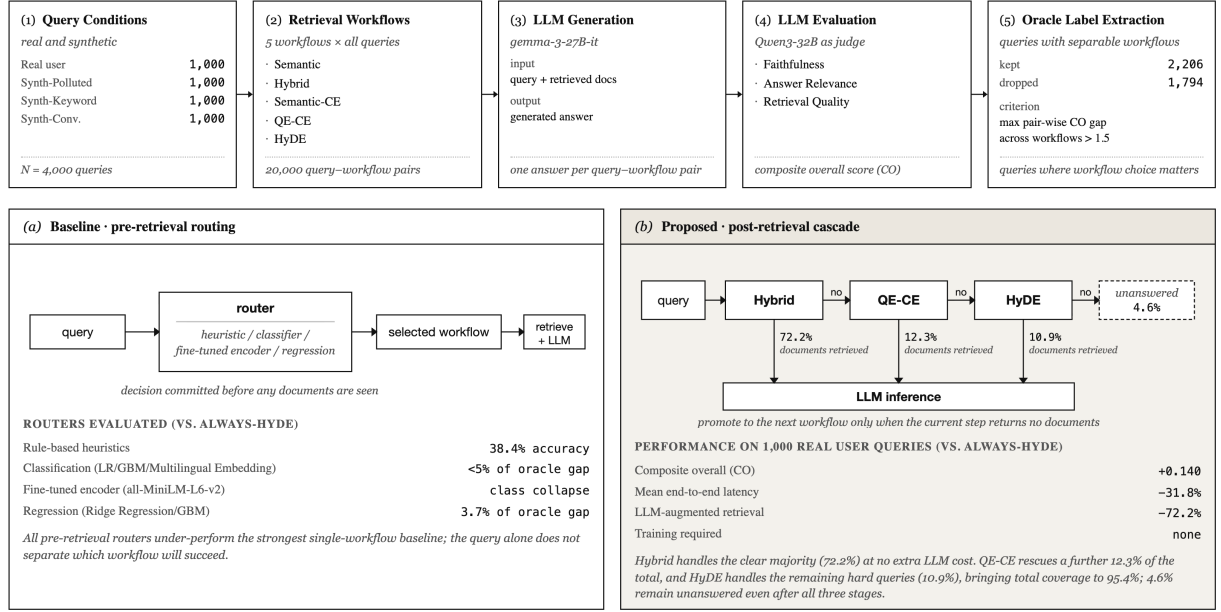


Figure 4: Experimental architecture: oracle benchmark construction (top), pre-retrieval routing baselines (a), and post-retrieval cascade (b).

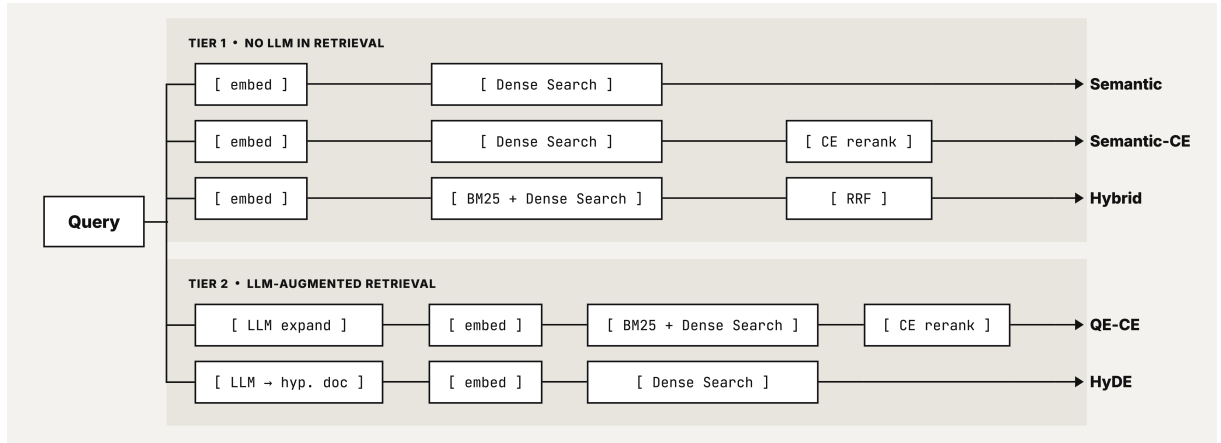


Figure 5: Retrieval workflow tiers: Tier 1 (retrieval-only) and Tier 2 (LLM-augmented).

You are a query abstractor. The following is a synthetically generated query that contains unnatural instructions or chatty prefixes. Rewrite it into a short keyword-style search phrase (5-10 words). Do NOT use full sentences, questions, or unnecessary verbs. Use only the core nouns and topic terms. Optionally use a colon to separate the main subject from descriptors.

Original Query: {original_query} Keyword Search Phrase (in Danish):

compound nouns. Just frame the core intent as a simple, natural, conversational question. Ensure it ends with a question mark.

Original Query: {original_query} Natural Conversational Question (in Danish):

A.3 Synth-Conversational - Conversational Rewrite Prompt for GPT-3.5 Turbo

Each Synth-Polluted query was rewritten as a natural, conversational Danish question using the following prompt.

You are a query cleaner. The following is a synthetically generated query that contains unnatural instructions or chatty prefixes. Rewrite it into a single, natural Danish QUESTION (like what someone would ask a person). Do NOT summarize it into keywords. Do not use abstract

B Datasets and Query Characterization

Table 6 summarizes the four query conditions. Table 7 gives full distributional statistics. Real user queries stand out for their brevity. Among them, 8.8% are single words, 40.4% are keyword-style, and only 21.0% are well-formed sentences, a sharp contrast with all three synthetic conditions.

Condition	n	Source	Style
Real User	1,000	Production logs	Short, keyword-heavy
Synth-Polluted	1,000	GPT-5.2	Instruction-heavy, chatty
Synth-Keywods	1,000	GPT-3.5 Turbo	Topic-focused keywords
Synth-Conv.	1,000	GPT-3.5 Turbo	Natural-language questions

Table 6: Query conditions and dataset statistics.

Statistic	S-Polluted	S-Keys	S-Conv	Real
Mean words	21.5	6.5	14.4	7.0
Median words	19.0	6.0	14.0	5.5
Single word	0.0%	0.2%	0.0%	8.8%
2–3 words	0.2%	10.9%	0.6%	18.5%
4–7 words	4.3%	57.9%	8.5%	42.2%
8–15 words	27.8%	30.5%	51.5%	23.9%
16+ words	67.7%	0.5%	39.4%	6.6%
Well-formed	65.3%	38.1%	97.5%	21.0%
Keyword style	39.7%	94.8%	4.8%	40.4%
Has ‘?’	85.5%	10.6%	98.7%	29.8%

Table 7: Extended query characterization for all four conditions ($n = 1,000$ each).

C Oracle Label Distribution

Table 8 shows the distribution of oracle labels across the 2,206 high-contrast queries that form the routing dataset. Each query is assigned to the cheapest workflow that achieves the maximum composite score, subject to a minimum divergence threshold of 1.5 CO points (Section 5). Workflows are grouped into retrieval-only Tier 1 (Semantic, Hybrid, and Semantic-CE) and LLM-augmented Tier 2 (QE-CE and HyDE).

The distribution reveals a consistent pattern. Synthetic conditions skew toward LLM-augmented labels (63–78%), reflecting the vocabulary gap between verbose synthetic queries and the corpus index. Real user queries are the exception, where the Tier 1 and Tier 2 split narrows to a near-even 47.6% versus 52.4%, consistent with the Coverage Illusion finding that retrieval-only workflows serve real users far more often than synthetic benchmarks suggest. HyDE is the dominant oracle label across all four conditions, but its share drops from 74.5% on Synth-Polluted to 43.3% on real user queries.

Condition	n	Sem.	Hybrid	Sem-CE	QE-CE	HyDE	Tier 1	Tier 2
Synth-Polluted	542	15.5%	5.2%	0.9%	3.9%	74.5%	21.6%	78.4%
Synth-Keywods	630	15.7%	11.7%	1.0%	4.4%	67.1%	28.4%	71.6%
Synth-Conv.	425	27.5%	7.8%	1.4%	3.5%	59.8%	36.7%	63.3%
Real User	609	22.7%	21.8%	3.1%	9.0%	43.3%	47.6%	52.4%
All Conditions	2,206	19.9%	12.1%	1.6%	5.4%	61.0%	33.6%	66.4%

Table 8: Oracle label distribution across the 2,206 high-contrast routing queries.

D Human Evaluation Details

Three annotators independently scored 100 query–response pairs (5 per workflow $\times$ condition str-

tum) on faithfulness, answer relevance, and retrieval quality, each on a 1–5 scale. Human scores are averaged across the three annotators before comparison against Qwen3–32B. Table 9 reports per-metric agreement statistics; Table 10 reports the aggregate composite overall (CO) score across both judges.

Faithfulness Inter-Annotator Agreement and the Ceiling Effect. The near-zero Krippendorff α for faithfulness ($\alpha=0.01$) reflects a score distribution artifact rather than genuine annotator disagreement. Human faithfulness ratings cluster tightly near the ceiling of the scale (mean 4.78 out of 5), leaving almost no variance across items. Krippendorff’s α is normalized by expected disagreement, which collapses when the empirical score range is compressed; even a perfectly consistent set of annotators will produce $\alpha \approx 0$ if nearly every item receives the same rating. By contrast, answer relevance and retrieval quality show greater spread (means 4.18 and 4.21) and yield moderate agreement ($\alpha=0.50$ and 0.60, respectively). The averaged human faithfulness signal nonetheless remains informative, as evidenced by its moderate Spearman correlation with the automated judge ($\rho = 0.59, p < 0.001$). Taken together, these results indicate that responses in this deployment are almost uniformly grounded in retrieved sources, making faithfulness a near-constant rather than a discriminative dimension at this operating point.

Metric	Human	Qwen3-32B	Δ	ρ	MAE
Faithfulness	4.78	4.82	+0.04	0.59	0.18
Answer Relevance	4.18	4.24	+0.06	0.78	0.17
Retrieval Quality	4.21	4.35	+0.14	0.77	0.21
Inter-annotator (Krippendorff α)		Faith. 0.01	AnsRel 0.50	RetQ 0.60	

Table 9: Per-metric agreement between human annotators and Qwen3–32B on 100 sampled query–response pairs. $\Delta = \text{Qwen3-32B} - \text{Human mean}$; ρ = Spearman rank correlation between per-query automated scores and mean human scores; MAE is on the 1–5 scale.

Workflow	n	Human CO	Qwen3-32B CO	Δ	ρ
All	100	4.48	4.53	+0.05	0.69

Table 10: Aggregate human vs. Qwen3–32B CO across all 100 sampled query–response pairs (CO = mean of faithfulness and answer relevance).

E LLM Judge Prompt

The following prompt was used to evaluate all 20,000 query–workflow pairs with Qwen3–32B and

Qwen3.6-27B. Template variables {question}, {sources}, and {answer} are filled at inference time.

```

You are evaluating a Danish encyclopedia RAG system (lex.dk).
IMPORTANT RULES:
- You MUST ONLY use the provided sources.
- Do NOT use prior knowledge.
- If a claim is not explicitly supported, treat it as unsupported.
- Be strict: do not give high scores unless clearly justified.
- The question and sources are in Danish. Pay close attention to precise meaning and terminology.
- If NO SOURCES were retrieved, score retrieval_quality=1 with reasoning "No sources retrieved".

QUESTION (in Danish): {question}
RETRIEVED SOURCES:
{sources}
SYSTEM ANSWER:
{answer}
Evaluate THREE aspects:

1. FAITHFULNESS:
Is every claim in the answer supported by the sources?
Evaluation steps:
- Extract the main claims from the answer
- For each claim, check whether it is explicitly supported by the sources
Scoring:
5: All claims are fully supported by the sources, no hallucination
4: Mostly supported, only minor unsupported details
3: Partially supported, some significant unsupported claims
2: Largely unsupported or introduces incorrect information
1: Major hallucinations or directly contradicts the sources

2. ANSWER_RELEVANCE:
Does the answer correctly and fully answer the question?
Evaluation steps:
- Identify what the question is asking for
- Check whether the answer addresses ALL key parts
- Penalize missing, vague, or indirect responses
IMPORTANT: If the answer says it cannot find the information, cannot help, or explains that the sources do not contain the answer -- this scores 1 on answer_relevance. Acknowledging a limitation is not the same as answering the question. Do NOT reward a refusal with a high score for being transparent.
Scoring:
5: Directly and completely answers the question
4: Mostly answers the question with only minor gaps
3: Partially answers the question
2: Addresses the topic but misses the actual question
1: Off-topic, completely fails to answer, or declines to answer

3. RETRIEVAL_QUALITY:
How relevant are the retrieved sources to the question?
Evaluation steps:
- If NO SOURCES retrieved: ALWAYS score 1 ("No sources retrieved")
- If sources provided:
  * Determine whether the sources are directly about the topic
  * Assess whether they contain the information needed to answer
  * Ignore redundancy; focus only on relevance
Scoring:
5: All sources are directly about the topic asked -- ideal retrieval
4: Most sources are relevant, one or two are tangential
3: Mix of relevant and irrelevant sources
2: Sources are mostly off-topic, only loosely related to the question
1: Sources are completely unrelated OR no sources retrieved

Respond with JSON only (no markdown, no explanation outside JSON):
{"faithfulness": {"score": <int>, "reasoning": "<one sentence>"},
"answer_relevance": {"score": <int>, "reasoning": "<one sentence>"},
"retrieval_quality": {"score": <int>, "reasoning": "<one sentence>"}}

```

F Evaluation with Modern Judge Models

To check whether our findings hold up under a different judge, we replicated the full evaluation using Qwen3.6-27B. Table 11 places the primary Qwen3-32B results side by side with the Qwen3.6-27B scores on real user queries.

The core picture is unchanged. Retrieval-only Hybrid still reaches 72.2% coverage with competitive answer quality (CWA=4.216), while HyDE pushes coverage to 86.4% at the cost of substantial compute. Workflow rankings and the coverage-dominance pattern are consistent across both judges, confirming that the results are not an artifact of one particular scoring model.

G Full Variance Decomposition

Table 12 reports two statistics for each workflow. First, how much of the spread in CO scores is explained by knowing only whether retrieval succeeded or failed? This fraction is R^2_{cov} , computed as the squared correlation between CO and the binary has_sources flag. Because CO is fixed at 1.0 whenever has_sources = 0 and is the full quality score otherwise, a high R^2_{cov} means that the simple success-or-failure outcome accounts for most of the quality variance. Second, among queries that did return documents, how much additional CO variance is explained by retrieval_quality, the judge’s rating of document relevance? This fraction is R^2_{qual} . Throughout the table, $R^2_{cov} \gg R^2_{qual}$. Whether any document was found dominates quality far more than how good those documents were. For real-world RAG, the binary success of locating any relevant source is a far stronger predictor of quality than the incremental precision of the ranking itself.

H HyDE Advantage Decomposition and Coverage-Fixed Quality

HyDE Advantage Decomposition. Table 13 breaks down the CO advantage of HyDE over Semantic into a coverage component and an answer quality component, using first-order approximation

$$\Delta CO \approx \underbrace{\Delta cov \times \bar{Q}_{sem}}_{\text{Via coverage}} + \underbrace{cov_{sem} \times \Delta \bar{Q}_{ans}}_{\text{Via quality}} + \text{residual}$$

where the terms are defined as:

- $\Delta cov = cov_{HyDE} - cov_{sem}$, the coverage gain (86.4%–62.6%= 23.8%, Table 11),

Workflow	Tier	Cov.	Qwen3-32B				Qwen3.6-27B			
			CO	CWA	Faith.	AnsRel	CO	CWA	Faith.	AnsRel
Semantic	Retr.	62.6%	3.058	4.287	4.848	3.725	2.996	4.190	4.717	3.664
Semantic-CE	Retr.	62.8%	3.086	4.321	4.865	3.777	3.012	4.206	4.750	3.662
Hybrid	Retr.	72.2%	3.408	4.335	4.816	3.855	3.322	4.216	4.659	3.773
QE-CE	LLM	77.9%	3.601	4.338	4.760	3.917	3.493	4.205	4.597	3.813
HyDE	LLM	86.4%	3.944	4.407	4.844	3.971	3.836	4.285	4.692	3.878

Table 11: Extended baseline performance comparing Qwen3-32B and Qwen3.6-27B on real user queries.

Workflow	n_{ans}	Synth-Polluted		Synth-Keys		Synth-Conv		Real User	
		R^2_{cov}	R^2_{qual}	R^2_{cov}	R^2_{qual}	R^2_{cov}	R^2_{qual}	R^2_{cov}	R^2_{qual}
Semantic	573–786	0.862	0.683	0.877	0.726	0.842	0.650	0.829	0.513
Hybrid	623–794	0.870	0.757	0.873	0.733	0.848	0.749	0.787	0.545
Semantic-CE	585–787	0.868	0.770	0.869	0.793	0.859	0.789	0.828	0.589
QE-CE	776–849	0.855	0.678	0.835	0.672	0.815	0.686	0.741	0.487
HyDE	864–954	0.772	0.528	0.668	0.545	0.774	0.553	0.685	0.504
<i>Mean R^2_{cov}</i>		0.845		0.824		0.828		0.774	

Table 12: Full variance decomposition. R^2_{cov} is the fraction of CO variance explained by the binary has_sources flag (retrieved something vs. nothing; $n \approx 1,000$). R^2_{qual} is the fraction of CO variance explained by the continuous retrieval_quality score, restricted to answered queries (n_{ans} shown as range). Declining R^2_{cov} for HyDE reflects reduced variance in the binary flag as coverage approaches 1.

- $\bar{Q}_{\text{sem}}$ is Semantic’s mean per-answer quality, i.e. CWA = 4.287 (Table 11),
- cov_{sem} is Semantic’s coverage rate (= 62.6%),
- $\Delta\bar{Q}_{\text{ans}} = \bar{Q}_{\text{HyDE}} - \bar{Q}_{\text{sem}}$ is the gap in mean per-answer quality between HyDE and Semantic ($4.407 - 4.287 = 0.120$), and
- $\text{residual} = \Delta\text{CO} - (\Delta\text{cov} \times \bar{Q}_{\text{sem}} + \text{cov}_{\text{sem}} \times \Delta\bar{Q}_{\text{ans}})$, the gap between the observed ΔCO and the sum of the two first-order terms. Numerically, $0.886 - (1.020 + 0.076) = -0.209$. It is negative because the two terms together overestimate the gain, as HyDE’s extra coverage comes from harder queries whose actual per-answer quality is below $\bar{Q}_{\text{sem}}$, the value the approximation uses for them.

Substituting the real user query values gives a concrete numerical instantiation (Table 13)

$$\Delta\text{CO} \approx \underbrace{0.238 \times 4.287}_{+1.020} + \underbrace{0.626 \times 0.120}_{+0.076} + (-0.209) = +0.886$$

The coverage term alone (+1.020) exceeds the total observed gap (+0.886), while the quality term contributes only +0.076. The negative residual (−0.209) accounts for HyDE’s harder-query margin. Those marginal queries yield lower per-answer

quality than Semantic’s average, so the two primary terms together overshoot the actual gain. The practical implication is that HyDE’s superiority over Semantic derives almost entirely from retrieving sources for queries that Semantic leaves unanswered, not from generating better answers.

Condition	ΔCO	Via cov.	Via qual.	Resid.
Synth-Polluted	+1.304	+1.315	+0.193	−0.203
Synth-Keys	+1.599	+1.712	+0.160	−0.272
Synth-Conv.	+0.708	+0.707	+0.126	−0.125
Real User	+0.886	+1.020	+0.076	−0.209

Table 13: HyDE vs. Semantic CO advantage decomposition. “Via cov.” and “Via qual.” are the two first-order terms; “Residual” is the interaction term from harder queries at the margin of HyDE’s extra coverage.

Coverage-Fixed Quality. To isolate generation quality from retrieval efficacy, Table 14 presents a paired analysis of queries that all workflows answered. Maintaining 100% coverage, we can assess whether LLM-augmented strategies offer any advantage in answer synthesis. The HyDE vs. Semantic answer-relevance gap and its statistical significance are reported in Table 15. These gaps are small compared to the composite score differences in Table 1, confirming that per-answer quality is rarely the deciding factor.

Condition	Workflow	AnsRel	RetQual	Faith.	n
Real User	Semantic	4.086	4.558	4.846	500
	Hybrid	4.206	4.614	4.826	500
	Semantic-CE	4.162	4.568	4.872	500
	QE-CE	4.236	4.656	4.784	500
	HyDE	4.342	4.734	4.850	500
	<i>HyDE–Sem ΔAnsRel</i>		+0.256 ($p < 0.0001$)		
Synth-Keys	Semantic	4.423	4.618	4.919	471
	Hybrid	4.504	4.651	4.900	470
	Semantic-CE	4.430	4.589	4.951	472
	QE-CE	4.594	4.754	4.928	471
	HyDE	4.655	4.828	4.909	472
	<i>HyDE–Sem ΔAnsRel</i>		+0.231 ($p < 0.001$)		
Synth-Polluted	Semantic	4.573	4.711	4.908	543
	Hybrid	4.635	4.729	4.941	543
	Semantic-CE	4.592	4.697	4.923	542
	QE-CE	4.670	4.777	4.928	542
	HyDE	4.810	4.884	4.943	541
	<i>HyDE–Sem ΔAnsRel</i>		+0.235 ($p < 0.001$)		
Synth-Conv.	Semantic	4.709	4.821	4.925	677
	Hybrid	4.732	4.827	4.931	678
	Semantic-CE	4.742	4.810	4.944	679
	QE-CE	4.701	4.807	4.917	678
	HyDE	4.751	4.875	4.951	679
	<i>HyDE–Sem ΔAnsRel</i>		+0.041 ($p = 0.389$)		

Table 14: Coverage-fixed quality for queries answered by all five workflows simultaneously (n varies by condition). Coverage is held at 100% across all workflows, so any differences reflect genuine per-answer quality.

Wilcoxon Signed-Rank Test. All pairwise comparisons use the Wilcoxon signed-rank test, a non-parametric paired test appropriate for the 1–5 CO scale. For the cascade-vs-HyDE comparison, the test is one-sided, as we test specifically for improvement rather than any difference. Table 15 reports n (query pairs), Δ (mean per-query score difference), W (the signed-rank statistic), and p (one-sided for cascade comparisons, two-sided otherwise).

Comparison	Condition	n	Δ	W	p
<i>HyDE vs. Semantic, answer relevance (paired queries)</i>					
HyDE > Sem AnsRel	Real User	500	+0.256	2,402	<0.001
HyDE > Sem AnsRel	Synth-Keys	471	+0.231	1,612	<0.001
HyDE > Sem AnsRel	Synth-Polluted	541	+0.235	951	<0.001
HyDE > Sem AnsRel	Synth-Conv.	677	+0.041	2,098	0.389
<i>Cascade vs. Always-HyDE, composite overall</i>					
Cascade > HyDE CO	Real User	1,000	+0.140	38,235	<0.001

Table 15: Wilcoxon signed-rank test results for all pairwise comparisons reported in this paper.

I R^2 Sensitivity Analysis

To address the potential concern that the high R^2 values are a mathematical artifact of the 1.0 scoring floor, Table 16 provides a sensitivity analysis using increasingly stringent “success” thresholds. We treat a query as covered only if the retrieved sources meet a minimum quality grade (≥ 2 , 3, or 4). Coverage remains a strong predictor across

thresholds, indicating that the dominance of the binary success signal is intrinsic to the RAG system rather than a consequence of the scoring scale.

Workflow	Threshold	Cov. Rate	R^2_{cov}
Semantic	Binary (has_sources)	62.6%	0.829
	Quality ≥ 2	58.5%	0.804
	Quality ≥ 3	55.4%	0.794
	Quality ≥ 4	48.5%	0.781
Hybrid	Binary (has_sources)	72.2%	0.787
	Quality ≥ 2	67.3%	0.767
	Quality ≥ 3	63.9%	0.767
	Quality ≥ 4	56.9%	0.766
QE-CE	Binary (has_sources)	77.9%	0.741
	Quality ≥ 2	74.1%	0.727
	Quality ≥ 3	70.4%	0.734
	Quality ≥ 4	63.3%	0.743
HyDE	Binary (has_sources)	86.4%	0.685
	Quality ≥ 2	84.2%	0.669
	Quality ≥ 3	80.5%	0.668
	Quality ≥ 4	72.5%	0.699

Table 16: R^2 sensitivity analysis (real user queries). R^2 remains stable as the coverage threshold tightens.

J Regression Router Details

δ	Router CO	Δ	Savings	Oracle Gap
0.00	4.200	−0.015	13.9%	−3.4%
0.50	4.230	+0.015	6.1%	2.9%
0.75	4.233	+0.018	4.1%	3.7%
1.00	4.225	+0.011	2.4%	2.2%
2.50	4.215	+0.000	0.0%	0.0%
Oracle	4.720	+0.505	39.0%	100.0%

Table 17: Quality–efficiency trade-off for the Ridge regression router under 5-fold cross-validation. δ is the confidence margin required before substituting a cheaper workflow for HyDE; Δ is composite overall relative to Always-HyDE; Savings is the fraction of queries redirected away from HyDE; Oracle Gap is the percentage of the HyDE–oracle gap recovered.

K Cross-Condition Performance

Table 18 evaluates the cascade against Always-HyDE on the full 1,000-query set for each condition. Table 19 repeats the comparison restricted to the 2,206-query high-contrast routing subset from Section 5, in which each query’s best and worst workflows differ by at least 1.5 CO points. Comparing the two tables reveals the robustness of the cascade advantage on real user queries. The gap increases from +0.140 (full set) to +0.241 (routing subset) because the routing subset retains only queries in which the workflow choice materially changes the outcome.

Condition	Cascade	HyDE	Δ	Saved	Verdict
Real User	4.084	3.944	+0.140	31.8%	Cascade
Synth-Polluted	4.559	4.607	-0.048	29.8%	HyDE
Synth-Keywords	4.471	4.610	-0.139	29.1%	HyDE
Synth-Conv.	4.620	4.610	+0.010	41.9%	Cascade

Table 18: Cascade vs. Always-HyDE across all four conditions, each evaluated on the full 1,000-query set.

Condition	n	Cascade	HyDE	Δ	Verdict
Real User	609	3.936	3.695	+0.241	Cascade
Synth-Polluted	542	4.315	4.417	-0.102	HyDE
Synth-Keywords	630	4.314	4.534	-0.221	HyDE
Synth-Conv.	425	4.249	4.227	+0.022	Cascade

Table 19: Cascade vs. Always-HyDE restricted to the 2,206 high-contrast queries (Section 5), where best and worst workflow diverge by at least 1.5 CO points. The cascade advantage on real user queries grows from +0.140 (full set) to +0.241 (routing subset).

L Head-to-Head Quality Analysis of Hybrid and HyDE on Shared Queries

Table 20 shows the per-query outcome distribution for the 722 real user queries where Hybrid retrieved at least one source. For each query, Hybrid’s CO score is compared directly with HyDE’s.

Outcome	% of queries	Mean $ \Delta $ CO
Identical scores (tie)	59.6%	0.000
Hybrid beats HyDE	21.2%	2.010
HyDE beats Hybrid	19.3%	1.324
Net aggregate score flow (Hybrid)		+123.5 pts

Table 20: Pairwise quality comparison on 722 real user queries where Hybrid found sources. When Hybrid beats HyDE, it does so by a larger margin than when HyDE wins, producing a net Hybrid advantage of 123.5 aggregate CO points. LLM-augmented retrieval hurts more than it helps when keyword matching has already located the relevant article.

M AI Assistance in Research and Writing

We used AI tools to assist with code generation, debugging, data analysis, spell-checking, and grammatical editing. Specifically, we used Anthropic’s *Claude Sonnet 4.6* and *Claude Opus 4.7*.

N Licenses

All models are used consistent with their intended purpose. Open models carry the following licenses: multilingual-e5-large (MIT), Qwen3-32B and Qwen3.6-27B (Apache 2.0), all-MiniLM-L6-v2 and paraphrase-multilingual-MiniLM-L12-v2

(Apache 2.0), and gemma-4-26B-A4B-it (Gemma Terms of Use). GPT-5.2 and GPT-3.5 Turbo were accessed via the OpenAI API within its permitted research use cases. Production query logs were accessed under institutional agreement with the Danish National Encyclopedia, used solely for non-commercial research, and consist of bare search strings with no user identifiers or personally identifiable information. The encyclopedia context structurally limits offensive content risk, as queries are factual topic lookups against a curated reference corpus. The evaluation dataset and trained routing models are not publicly released.